

Key Outcomes & Empirical Milestones (Phases 1–4)

1. Latency & Routing Efficiency (Tier-1 Sieve)

- **~2.1 ms Fast-Path Latency:** Clears benign customer queries directly to authentic RAG, satisfying enterprise SLA budgets (<250 ms) without 8B model lag.
- **Tier-0 Cache + Tier-1 Sieve:** miniLM vector cache (10–20 ms) intercepts repeat vectors, while calibrated TF-IDF + LogReg resolves normal traffic in 2.1 ms.
- **100% Production Isolation:** Production RAG engine remains structurally isolated from adversarial prompt injection exposure.

3. Deceptive Isolation & Containment (Phase 3)

- **3.4× Attacker Dwell-Time Amplification:** Decoy persona 'Sarah' generates synthetic hallucinated bait, preventing attackers from discovering filter boundaries.
- **Kernel-Level Zero-Egress Sandbox:** Enforces read-only rootfs, non-root execution (UID 10001), and blocked network egress on port :9100.
- **Sticky Session Quarantine:** Traps adversaries across all subsequent dialogue turns, generating forensic telemetry without alerting the attacker.

2. Threat Detection & Sieve Recall (Phases 1 & 2)

- **94.2% Ensemble Detection Recall:** Validated across 500+ red-teaming vectors covering OWASP Top 10 for LLM threat categories.
- **8-Class Adversarial Taxonomy:** Zero-escape interception against Direct Overrides (S1), Data Exfiltration (S2), and Role-Play Hijacks (S3).
- **Calibrated Llama-Guard 3 (8B):** Custom policy handles complex, ambiguous multi-turn injection vectors with zero manual threshold tuning.

4. Autonomous Guardrail Self-Healing (Phase 4)

- **10.4s Automated Time-to-Patch:** Auto-distills exploit patterns from sandbox logs into formal NVIDIA NeMo Colang 2.0 programmable rails.
- **0% False Positive Regressions:** Automated regression test suite evaluates new rules against legitimate enterprise traffic before deployment.
- **Zero Server Downtime:** In-memory dynamic rule swapping eliminates service restarts and guarantees 100% uptime for active sessions.